ClassX srl
via Del Tiglio, 91 I-56012 Calcinaia - Montecchio (PI) – ITALY
Tel. +39.0587.488128 Fax +39.0587.488128 - C.F. - P.I. 01351840507
Email: info@classx.it HomePage: www.classx.it

StealthCG™

Protocol Specifications
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1 How to use the StealthCG protocol

StealthCG™ is a remote graphics engine that allows to interact with many graphic contents according to a set of commands. The contents are generally graphics templates created by MoreCG™ or CastaliaCG™ by ClassX, but they also can be plugins following the ClassX MediaPlugs™ architecture.

StealthCG integrates all the ClassX advanced technologies designed to get the best performance and flexibility: SGL™, FrameLinkDrivers™, DataSource™ and MediaPlugs™. Please read the related documentations for more information.

StealthCG supports many different kinds of graphic contents as tickers, clocks, timers, MoreCG and CastaliaCG graphic templates. The graphics engine paints each content into a different layer, according to its 'visual priority' field value (default is 50). See MediaPlugs documentation for further details.

Each MediaPlug is identified by a type according with the following MediaPlug **PlugIn table** returned by the command GETCONTENTSTYPES

TYPE	MediaPlug
PLUGIN_LOGO	Logo
PLUGIN_BACKGROUND	Background
PLUGIN_TICKERH	Horizontal ticker
PLUGIN_SIMPLEMEDIAPLAYER	Simple Media Player
PLUGIN_MEDIAPLAYER	MediaPlayer
PLUGIN_LIVECLOCK	LiveClock
PLUGIN_LIVEDIGICLOCK	LiveDigiClock
PLUGIN_LIVETIMER	LiveTimer
PLUGIN_MEDIASHAPE	MediaShape
PLUGIN_SLIDESHOW	SlideShow
PLUGIN_TEXTSLIDESHOW	TextSlideShow
PLUGIN_SMSTICKERH	SMSTicker
PLUGIN_TEXTAREA	TextArea
PLUGIN_MEDIAKARAOKE	MediaKaraoke
PLUGIN_STYLEDTICKERH	StyledTicker

See MediaPlugs™ user manual to get more information about their functionality.

1.1 Connecting to StealthCG

The users can interact with StealthCG through a TCP/IP port (the COMMUNICATION_PORT, default is 101). StealthCG processes each command sequentially and always returns either the command execution result or an error code in case the command fails. Each command is processed immediately.

StealthCG supports multiple clients connections and integrates the most advanced technologies to synchronize the commands execution.

The users are provided a second TCP/IP port (the EVENT_PORT, default is 201) that is used by StealthCG to send asynchronous events and notifications to the user application. This port is output only and does not accept any command. Users connect to this port to have fine-grained control over the StealthCG contents execution i.e. to synchronize the execution of a number contents.

Users can set their preferred ports in the StealthCG preference panel.

A common client application connects to the COMMUNICATION_PORT to control StealthCG and optionally to EVENT_PORT to receive asynchronous event messages and notifications.

1.2 Special characters

All sent and received protocol strings are in ASCII format and are always closed by quote chars. They may contain special chars in ANSI-C format (escape sequences) as line feed, carriage return etc. The following table contains all supported special characters:

NEWLINE	\n
HORIZONTAL TAB	\t
BACKSPACE	\b
RETURN	\r
FORM FEED	\f
BACKSLASH	\\
SINGLE QUOTE	\'
DOUBLE QUOTE	\"

Example:
>**GETPROJECTOBJECTS**
<OBJECTNAME="local",VALUE="Restaurant \"The Boat\" Milan",KIND=TEXT

In this case the **value** of the **local** object is **Restaurant "The Boat" Milan**

In case of file path string this protocol supports both Back Slash (DOS/Windows) and Slash (Unix/Linux/Mac) characters.

NOTE: The protocol is always parsed line by line and each line sent to or received from the protocol handler is always terminated by a **newline** character. It is always possible to include multiple newlines characters in a command line by replacing any occurrence with the "\n" sequence.

>**GETPROJECTOBJECTS**
<OBJECTNAME="local",VALUE="Restaurant\nMilan",KIND=TEXT

the VALUE will be parsed as:

Restaurant
Milan

1.3 Commands returning lists

Some of the commands in the protocol can return a list of values. The lists are sent to the calling application by a sequence of Line-feed ('\r') terminated lines.

1.4 Content identification

Each content is identified uniquely by a string 'ID' that will be used as a unique key to detect the content and interact with it. The content ID is not case sensitive and cannot be null (i.e. “”). Any leading or trailing white spaces are removed automatically by the StealthCG protocol handler. The content ID may contain only [a..z], [A..Z][0..9]["-","_"] characters.

1.5 The Commands syntax parsing

The TCP/IP protocol parser is based on the ArgParser by ClassX, a specialized set of classes that allows flexible parsing of the commands sent to the protocol handler.

The ArgParser will parse the built-in command templates for each command of the protocol.

The **Help** command gives a list of the entire set of commands allowed by the protocol. Giving a closer look at the list, we can see that almost any command is followed by a series of parameters. Each parameter is marked with a control character sequence that rules the interpretation of the command:

PARAMETER Marker	Interpretation
<no sequence>	Normal parameter
\k	Keyword parameter
\r	Required parameter
\s	Swithch parameter (TRUE or FALSE)
[...]	Param information: ignored when parsing

The ArgParser will process each command by the following algoritm:

1. Discard param information like square bracket suggestions.
2. Search and assign the Required parameters. If one or more Required parameters is missing, then there will be a parsing error.
3. Search and assign the Keyword parameters.
4. Search and assign the Switch parameters.
5. Assign the remaining arguments to the Normal parameters.

An example: the following command template

```
"COMMAND/r/k,PARAM0/r,PARAM1[A|B|C]/r,PARAM2/k,SWITCH/s"
```

the above means that

```
COMMAND is a keyword and is required
PARAM0,PARAM1 are required but not a keyword
PARAM2 is a keyword
SWITCH is a true/false switch
```

will parse the following argument string

```
COMMAND=comando PARAM0=-123.453, \"PIPPO \\\"LA\\\" SCOPA\", PARAM2=\"This is Param2\" , SWITCH=TRUE"
```

as follows:

```
COMMAND = comando
PARAM0 = -123.453
PARAM1 = PIPPO "LA" SCOPA
PARAM2 = This is PARAM2
SWITCH = true
```

Note that arguments are normally a comma separated list, but also spaces are allowed. Also complex control char sequences are supported. Named arguments always have the precedence in the value assignment:

```
"ARG0/k,ARG1\k"
```

will parse the following argument string

```
"ARG1=ciao,ARG0=pippo"
```

as follows:

ARG0=pippo
ARG1=ciao

and will parse the following argument string, without argument names

```
"ciao,pippo"
```

as follows:

```
ARG0=ciao
ARG1=pippo
```

More generally the sequence of the arguments is always interpreted left to right, except that "named" arguments (keywords) have the precedence.

The **Switch** argument must be specified by assigning its name to a "true" or a "false" value. It can also be specified by simply passing the argument name:

"DISPLAY\s\k" will parse the argument "DISPLAY=FALSE" as false, "DISPLAY=TRUE" as true, "DISPLAY" as true, "" as false.

1.6 Note

ATTENTION: All obsolete commands are tagged as **{DEPRECATED}**. These commands may be not supported by future protocol releases. Please avoid to use them.

2 Content and System commands

HELP

Return the list of the supported commands.

LOG <source, text>

Write a text into the LOG file. This may be usefull to add comments into the LOG file. Source tell who writes the comment

QUIT

Close the application.

CLEAR

Stop and remove all contents.

PING

Check if StealthCG is up and running. Always returns OK.

GETCONTENTS

Return a list of all contents with the syntax ID,TYPE,DESCRIPTION. The TYPE field identifies the content between TEMPLATE, PLUGIN and future content types. See the **PlugIn table**.

Example:

```
>GETCONTENTS
<"NEWS","TEMPLATE","\morecg\project\news.mcg"
"SCROLL NEWS","PLUGIN_livesuite.ticker.",""
"LOGO_APP","PLUGIN_it.classx.gfb.Logo","\shared\picture\images\classx.tga"
```

GETCONTENTTYPE <ID>

Return the content type between TEMPLATE, PLUGIN and future contents following the above PlugIn table.

Example:

```
>GETCONTENTTYPE "NEWS"

or

>GETCONTENTTYPE ID="NEWS"

<"TEMPLATE"
```

GETCONTENTSTYPES <ID>

Return a list of all available contents types.

Example:

```
>GETCONTENTSTYPES
<"PLUGIN_BACKGROUND"
 "PLUGIN_LIVECLOCK"
 "PLUGIN_LIVEDIGICLOCK"
 "PLUGIN_LIVETIMER"
 "PLUGIN_LOGO"
 "PLUGIN_MEDIAKARAOKE"
 "PLUGIN_MEDIAPLAYER"
 "PLUGIN_MEDIASHAPE"
 "PLUGIN_SIMPLEMEDIAPLAYER"
 "PLUGIN_SLIDESHOW"
 "PLUGIN_SMSTICKERH"
 "PLUGIN_TEXTAREA"
 "PLUGIN_TEXTSLIDESHOW"
 "PLUGIN_TICKERH"
 "TEMPLATE"
```

PLAYCONTENT <ID>

Play the content with the given ID. If the content is paused this command will resume it.

NOTE: Not all contents may be paused. It depends on their implementation. Generally only MoreCG, CastaliaCG templates can be paused.

PLAYCONTENTS <IDLIST,STOPLIST>

Stop and play a list of contents synchronously: IDLIST contains the content to play, and STOPLIST contains the contents to stop. In both cases the list must follow the syntax: IDLIST="ID1,ID2,ID3".
If one or more content IDs in the lists is not valid, the command will not be executed.

Example: play the content news,digital clock,ticker at the same time.

```
>PLAYCONTENTS "news,digital clock,ticker"

or

>PLAYCONTENTS IDLIST="news,digital clock,ticker"
```

Example: Stop the background and timer before, then play the content news,digital clock,ticker at the same time.

```
>PLAYCONTENTS IDLIST="news,digital clock,ticker", STOPLIST="background,timer"
```

Example: Stop and play the same content background. Since the operations are synchronous, you will not see any void frame because the background is stopped and played in the same frame.

```
>PLAYCONTENTS IDLIST="background" STOPLIST="background"
```

STOPCONTENT <ID>

Stop the content with the given ID.

STOPALL

Stop all playing contents

STOPCONTENTS <IDLIST>

Stop the given list of contents synchronously. The list must follow the syntax: IDLIST="ID1,ID2,ID3". If one or more content IDs in the list is not valid, the command will not be executed.

Example: stop the content news,digital clock,ticker at the same time.

```
>STOPCONTENTS "news,digital clock,ticker"

or

>STOPCONTENTS IDLIST="news,digital clock,ticker"
```

REMOVECONTENT <ID>

Stop and remove the content with the given ID.

SETCONTENTVISUALPRIORITY <ID>,<VALUE>

The StealthCG engine paint each content into a different layer identified by a number (default is 50). This command allows to change the visibility order of the content: 0 is the lower layer and 9999 is the higher layer.

GETCONTENTCOMMANDS <ID>

Return a list of all content commands. Each content can have a set of custom commands to control many internal aspects, behaviors and settings. Normally only the MediaPlugs expose this feature. For more information please consult the **MediaPlugs Commands** section.

Example: return a list of all specific commands of the ticker content.

```
>GETCONTENTCOMMANDS TICKER
<ADDPLAYLISTFILE FILEPATH/r
 SETVISUALPRIORITY PRI/k
 ADDTICKERTEXT TEXT/r,LOOPS
 RESETPLAYLIST
 CLEARPLAYLIST IMMEDIATELY/s
 SETPOSITION X/k/r,Y/k/r
 SETSIZE WIDTH/k/r,HEIGHT/k/r
```

SENDCONTENTCOMMAND <ID>,<COMMAND>

Send a command to the content with the given ID.
Each content can have a set of custom commands to control many internal aspects, behaviors and settings. Normally only the MediaPlugs expose this feature. For more information please consult the **MediaPlugs Commands** section.

To obtain a correct interpretation of the command parameter, it should be written by replacing special characters with the equivalent ANSI-C escape sequence. See the **Special Characters** table.

Example: change the image file of a Background plugin.

```
>SENDCONTENTCOMMAND ID="BGND", COMMAND="SetFilename \"C:\\\\Desktop\\\\image.jpg\""
```

GETCONTENTSTATUS <ID>

Return the current content status.

Return values:
- 0 the content is PLAYING
- 1 the content is PAUSED
- 2 the content is STOPPED

ADDPLUGIN <ID>,<TYPE>

Add the plugIn with the given TYPE following the above **PlugIn table**.

ADDSERIALIZEDPLUGIN <ID>,<VALUE>

Add a plugIn instance according to the plugIn type contained into the VALUE serialization string.
The VALUE may refers to a file that contains the serialized plugIn.

How to serialize a plugIn:
- Open LiveBoard or LiveBoard_SCG.
- Create and customize a plugIn as you want.
- Move the plugIn button over the copy button: the plugIn is now serialized into the System clipboard.
- Open the notepad and paste the clipboard (CTRL-V).
- Save the file as text file.

```
>ADDSERIALIZEDPLUGIN "tixker", "C:\\ClassX_Application\...\serialized_plugIn.txt"
```

SETCAPTUREMODE <FORMAT[TGA|PNG|AVI|QT]>,<EXPORTDIR>,<BASENAME>

Each captured frame is stored into a default directory using a default formatted name:

{export.dir} / "basename" + "_" + "frameCounter"

The {**export.dir**} is an environment variable and refers as default to the directory named "export" under the current application directory.

The **baseName** is a string prefix applied to each file name. Default value is "unnamed".

FrameCounter means the current frame number and is automatically increased by the application. It is normally formatted like "00000000".

The user can also customize the export directory and the basename using the related parameters. If the customized path does not exist the application will try to use the default value.

Format means the file format. TGA or PNG for image file and AVI or QT (QuickTime) for video file. Some of these formats may not be supported.

NOTE: when the application captures the frames the output device will appear not synchronized like it was dropping frames.
The alpha information will be preserved if the exporter supports it.

WARNING: The application will be present the codec selector in case of AVI format. This operation will lock the application until user selection.

STARTCAPTURE

Start to capture frames.

STOPCAPTURE

Stop frames capturing.

RECLAIMMEMORY

Free memory resources.
WARNING: calling this command while graphics elements are playing may cause drop frames.

GETCONTENTPREVIEW <ID>,<FILENAME>

Save a preview of the content ID into the given filename. This command may be not supported by all content types.

LOADPROJECT <FILENAME>

Load the StealthCG project. This operation may require few seconds due to the graphics elements initialization.
Return "Ok" if the operation has finished without errors or one of these errors types: ERR_LOADPROJECT, SCG_ERR_INVALIDCONTENTID

Example:

```
-> loadprject C:\ClassX_Applications\stealthcg\project\news\notiziario_01.cgp
<- OK
```

SAVEPROJECT <FILENAME>, <REPLACE>

Save the StealthCG project.

FILENAME is required, the filename where save the project
REPLACE is optional, default is true. Se this arg TRUE if StealthCG can replace the FILENAME that just exist or false.

Return "Ok" if the operation has finished without errors or one of these errors types:

- ERR_FILEALREADYEXIST The file already exist
- ERR_INVALIDFILENAME The filename is invalid
- SCG_ERR_SAVEPROJECTFAILED If StealthCG fail to save the project

Example:

```
-> saveproject C:\ClassX_Applications\stealthcg\project\news\notiziario_01.cgp REPLACE="TRUE"
<- OK
```

3 Template Commands

ADDTEMPLATE <ID>,<FILENAME>,<SHARED>

Add a MoreCG, CastaliaCG or CoralCG graphic template according with the given file name.
If the file name is valid it returns OK when the template initialization is done. This may require few seconds depending by the complexity and the resources used by the graphic template. If shared flags is true the template will be shared among many contents, all pointing to the same template.

The template objects are not updated to the currently DataSource values. In this case we suggest to call the UPDATETEMPLATEDATASOURCE command to update the template objects.

GETTEMPLATEOBJECTS <ID>

Return a formatted list of all *editable objects* contained in the MoreCG or CastaliaCG graphic template with the given ID. The list has the following format:

OBJECTNAME=<name>,VALUE=<value>,KIND=[TEXT],DATAFORMATTER=<format>

The object KIND describes the object type as TEXT, IMAGE, AVI, QTVIDEO.
DATAFORMATTER contains a formatted string where the value is applied. This is supported only by Object with KIND=TEXT

```
>GETTEMPLATEOBJECTS
<OBJECTNAME="name",VALUE="Cristian",KIND=TEXT,"<html><b>Hi,     my     name     is     <ID="NAME"
DEFAULTVALUE="Fabio" KIND="TEXT"></b>"
 OBJECTNAME="surname",VALUE="Smith",KIND=TEXT
 OBJECTNAME="team",VALUE="gfx/milan.tga",KIND=IMAGE
```

EDITTEMPLATEOBJECT <ID>,<OBJECTNAME>,<VALUE>

Allows to modify the value all editable objects with the given name contained into the graphic template at the given ID. This command can be applied to all objects kind and the result depends on their implementation. For example this command changes the text value for a TEXT object and also set the file path for an IMAGE or VIDEO object.

NOTE: StealthCG does not check the correctness of the value. It means that if the file path of an image is wrong StealthCG does not return any error, but the IMAGE object will display an empty image.

Example

```
>EDITTEMPLATEOBJECT "news","name","Adam"
<OK

>EDITPROJECTOBJECT "soccer","team","gfx/milan.tga"
<OK

>EDITPROJECTOBJECT "soccer","logo","gfx/milan_logo.avi"
<OK
```

GETTEMPLATEINFO <ID>,<TOTFRAMES|CURRFRAME|DESCRIPTION|FILE|ICONFILE|FRAMEMARKER|POSTERFRAME|FPS|RESOLUTION]>

Return the graphic template information as total frames, number of objects, the current frame, the template description, the file name, the template icon file path, the pause frames, the poster frame, the template FPS and resolution.

The command result depends by the required property

TOTFRAMES: template total frames

```
>GETTEMPLATEINFO "news",TOTFRAMES
<250
```

CURRFRAME: the current drawn frame

```
>GETTEMPLATEINFO "news",CURRFRAME
<127
```

DESCRIPTION: the template description

```
>GETTEMPLATEINFO "news",DESCRIPTION
<"This project has been designed to display dynamic news"
```

FILE: the template file path

```
>GETTEMPLATEINFO "news",FILE
<"C:\Programmi\ClassX\MoreCG\Project\tg\news.mcg"
```

ICONFILE: the template icon file path. The icon is a TGA image with a low resolution (90x72) pixels

```
>GETTEMPLATEINFO "news",ICONFILE
<"C:\Programmi\ClassX\MoreCG\Project\tg\news.tga"
```

FRAMEMARKER: a list of frame markers returning for each elements the frame and the marker type.

```
>GETTEMPLATEINFO "news",FRAMEMARKER
<FRAME=10,TYPE="PAUSE"
 FRAME=40,TYPE="MARKIN"
 FRAME=70,TYPE="MARKOUT"
 FRAME=100,TYPE="PAUSE"
```

NOTE: Nowadays StealthCG supports only PAUSE frames type. Please read MoreCG and CastaliaCG user manual to get more information about the FrameMarker functionality.

POSTERFRAME: the graphic template interesting frame, used by MoreCG and CastaliaCG to compute the template icon.

```
>GETTEMPLATEINFO 0,POSTERFRAME
<200
```

FPS: the graphic template frame per second

```
>GETTEMPLATEINFO 0,FPS
<30
```

RESOLUTION: the graphic template resolution

```
>GETTEMPLATEINFO 0,RESOLUTION
<WIDTH=720,HEIGHT=576
```

SETTEMPLATESPEED <ID>,<VALUE>

Set the graphic template speed percentage [1.500]. This value can be set also the project playing.

```
>SETTEMPLATESPEED 0,300
<OK
```

SETTEMPLATELOOP <ID>,<VALUE>

Set the graphic template loops number: the default value is 1. Use -1 for infinite loops.

```
>SETTEMPLATELOOP 0,-1
<OK
```

UPDATETEMPLATEDATASOURCE <ID,ASYNCH>

Force to reload all external DataSource files joined with the graphic template.
If ASYNCH is true the DataSource reloading is dispatched to a separated process. During the reloading the event port will fire the DataSourceEvents: the DATASOURCE_INIT_END will be fired when the reloading is completed.

```
>UPDATETEMPLATEDATASOURCE 0
<OK
```

GETTEMPLATEOBJECTVALUE <ID>,<OBJECTNAME>

Returns the current value of the given editable object name contained into the graphic template at the given ID. This command can be applied to all objects kind and the result depends on their implementation. For example this command returns the text value for a TEXT object and also returns the file path for an IMAGE or VIDEO object.

```
>GETTEMPLATEOBJECTVALUE "news","name","Adam"
<"Adam"

>GETTEMPLATEOBJECTVALUE "soccer","team"
<"C:\ClassXApplication\picture\weather\static\raccolta3\sun.tga"
```

TEMPLATENEXTRECORD <CONTENTNAME>,<DATASOURCEID>,<STEPS>

This command allows to change the RecordID of some DataSources assigned to the CONTENTNAME. In particularly this moves the RecordID to the next record about STEPS. If steps is not specified it is considered equal to 1. When the next record reaches the last, it will restart from the beginning automatically. The DataSourceID identifies one or more DataSources

TEMPLATEPREVIOUSRECORD <CONTENTNAME>,<DATASOURCEID>,<STEPS>

This command allows to change the RecordID of some DataSources assigned to the CONTENTNAME. In particularly this moves the RecordID to the previous record about STEPS. If steps is not specified it is considered equal to 1. When the next record reaches the first, it will restart from the beginning automatically. The DataSourceID identifies one or more DataSources

TEMPLATESETRECORD <CONTENTNAME>,<DATASOURCEID>,<RECORD>

This command allows to change the RecordID of some DataSources assigned to the CONTENTNAME. In particularly this moves the RecordID to the given record position. The record is clamped to the DataSource records number. The DataSourceID identifies one or more DataSources.

TEMPLATEEXITFROMLOOP<CONTENTNAME>

A graphic template may loop in a FrameMarker range. The loops number may be fixed or infinite. Inboth cases this command force the template to exit from the loop.

SETTEMPLATEOBJECTVISIBLE<ID>,<OBJECTNAME>,<VISIBLE>

Turn on/off the drawing of the template object with the given name.

4 Content specific commands

Each content support specific commands depending by its implementation. A specific commands can be sent to a content as parameter of the SENDCONTENTCOMMAND command.
There are some common commands for all contents that allow to set the position, dimension and visibility order. The command GETCONTENTCOMMAND returns a list of all content specific commands (see above).

5 Messages sent from StealthCG

StealthCG is able to send messages to its clients telling informations about occurred events or content status: for example when a content reaches a FrameMarker, StealthCG sends a message that notifies the reached frame event. Please remember that the communication is one way, the users receive the event message and process it without send any answer or confirm to StealthCG.

PING

StealthCG automatically detects when the client is disconnected and then disposes all resources allocated to send the events messages. StealthCG sends at fixed time interval the PING command to check the connection status. This command can be ignored by the users.

SERVEREVENT <ID>,<EVENTTYPE>

This is low level event fired from the StealthCG engine when an operation has been done or an event is occurred, according with the following events table:

ID	DESCRIPTION
0	Content added
1	Content removed
2	Content suspended
3	Content resumed
4	Server stared [Internal]
5	Server stopped [Internal]
6	Server suspended [Internal]
7	Server resumed [Internal]
8	Content paused
9	Content resumed from pause
10	Content reset
11	Content removing [Internal]
12	Content suspending [Internal]
13	FrameMarker reached

Example: content news has been added to the StealthCG engine.

```
<SERVEREVENT ID="news", TYPE=0>
```

6 DataSourceEvent messages

StealthCG is able to send messages to its clients telling informations about the DataSource status.

ID	DESCRIPTION
DATASOURCE_FILE_CHANGED	Fired when the Datasource file is changed
DATASOURCE_CHANGED	Fired when the Datasource has been initialized or reloaded
RECORDID_CHANGED	Fired when the Datasource record ID is changed
DATASOURCE_INIT_ERROR	Fired when the Datasource initialize is failed, otherwise the Datasource fires a DATASOURCE_CHANGED event type
DATASOURCE_INIT_START	Fired when the Datasource initializing is started
DATASOURCE_INIT_END	Fired when the Datasource initializing is complete

Example: content news has been added to the StealthCG engine.

```
<DATASOURCEEVENT TYPE="DATASOURCE_INIT_START", ID="twitter"
```

7 Deprecated Commands

Nowadays none of the commands are deprecated.

8 MediaPlug commands

ClassX srl
Via del Tiglio, 91 – Villa Montecchio – 56012 Calcinaia (PI) – ITALY
tel: 0587/488128 Email: support@classx.it Web: www.classxcg.com

PlugIns™ commands list

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8.1 Common commands

Each plugIn supports a set of remote commands that allows to change its appearance, position and settings. The plugIns have some common properties as position, dimension described in this section, while others properties depend from their implementation and will be illustrated into the related chapters.

SETPOSITION <X>,<Y>

Move the plugIn to the screen location X,Y. The parameter with value equals to -1 will be ignored.

Example: move the content to X = 100 coordinate keeping the current Y coordinate.

```
>SETPOSITION 100,-1;
<OK
```

SETSIZE <WIDTH>,<HEIGHT>

Resize the plugIn to the given WIDTH and HEIGHT. The parameter with value equals to -1 will be ignored.

Example: set the content width to 100 keeping the current height.

```
>SETSIZE 100,-1;
<OK
```

{DEPRECATED} SETVISUALPRIORITY <VALUE>

Set the plugIn visual priority in a range value [0,10000]: 0 is the lower layer and 10000 is the higher layer.

NOTE: This command does not produce any result.

Opening "**Remote Control Tester**" in ClassX distribution and **connecting** to LiveBoard, it is possible to view the list of available commands for each Plugin.

Just insert this command in the "**Command**" area:

LBC_GETCONTENTCOMMANDS <value>

8.2 Logo

SETFILENAME <FILE>

Set logo file name.

Example
```
>SetFileName \"C:\\\\Pictures\\\\image.png\"
<OK
```

SETLOGOPROPERTY <PLAYSPEED>, <PLAYMODE>, <FIELDFLAGS>, <USEALPHA>, <FPS>

Set the Logo properties. You can set one or more properties in the same command:

PLAYSPEED speed value percentage over original speed, used to play the logo [0..200]. The default is 100 (the original speed);

PLAYMODE mode used to play the logo [0..9]. The default value is 3 [PLAY_LOOP].

- 0 = still first frame [PLAY_STILLFIRST];
- 1 = still last frame [PLAY_STILLLAST];
- 2 = play one time only [PLAY_ONCE];
- 3 = loop animation [PLAY_LOOP];
- 4 = reverse loop animation [PLAY_REVLOOP];
- 5 = ping-pong animation [PLAY_PINGPONG];
- 6 = play one time only automatically [PLAY_ONCE_AUTO];
- 7 = loop animation automatically [PLAY_LOOP_AUTO];
- 8 = reverse loop animation automatically [PLAY_REVLOOP_AUTO];
- 9 = ping-pong animation automatically [PLAY_PINGPONG_AUTO];

FIELDFLAGS field handling [0 = keep original interlace fields, 1 = de-interlace frame];

USEALPHA use the alpha information of the logo if present (default is TRUE);

FPS play speed of the animation expressed in frames per second.

Example:
```
>SetLogoPropery PLAYSPEED=50
<OK
```
Set the play speed of the Logo to the 50% of the normal speed.

```
> SetLogoPropery PLAYMODE=5,USEALPHA
<OK
```
Ping-pong animation and maintain the frames alpha value.

8.3 Timer

SETSTARTTIME <VALUE>

Set the Timer start time in seconds or milliseconds. To set it in seconds it is sufficient to give it a numeric value, or the same value followed by “s”, while to set it in milliseconds the numeric value must be followed by “m”.

Example 1: set the start time to 20 seconds.

```
>SetStartTime \"20\"
<OK
```

Example 2: set the start time to 20 seconds.

```
>SetStartTime \"20s\"
<OK
```

Example 3: set the start time to 20000 milliseconds.

```
>SetStartTime \"20m\"
<OKSETENDTIME <VALUE>
```

Set the Timer end time in seconds.

Example: set the end time to 60 seconds.

```
>SetEndTime 60
<OK
```

SETTIMERDIRECTION <[UP|DOWN]>

Set the timer direction: set UP for a timer and DOWN for a countdown

Example: setup the timer for a countdown

```
>SetTimerDirection DOWN
<OK
```

RESETTIMER

Reset the timer to start time seconds.

STARTTIMER

Start the timer count.

STOPTIMER

Stop the timer count

GETTIMEMILLIS

Return the current timer time in milliseconds

SETTIMERFORMAT

Set the time format of the timer. The only supported formats are the following:

- `"s:tt"` - seconds and milliseconds.
- `"ss:tt"` - seconds and milliseconds with 2 cyphers for the seconds always visible.
- `"s"` - seconds.
- `"ss"` - seconds with 2 cyphers always visible.
- `"sss"` -seconds with 3 cyphers always visible.
- `"m:ss"` - minutes and seconds.
- `"mm:ss"` - minutes and seconds with 2 cyphers for the minutes always visible.
- `"mmm:ss"` - minutes and seconds with 3 cyphers for the minutes always visible.

- "HH:mm:ss" - hours, minutes and seconds.
- "HH:mm" - hours and minutes.
- "HH:mm:ss:tt" - hours, minutes, seconds and tenths of second.

Example: setup the timer to display hours, minutes, seconds and tenths of second.

```
>SetTimerFormat \"HH:mm:ss:tt\"
<OK
```

8.4 Horizontal and Styled Ticker

ADDTICKERTEXT <TEXT>, <LOOPS>

Set the text to scroll into the Ticker defining also the loops numbers. Use –1 for infinite loop.

Example: set the text and loops to 3 times

```
>AddTickerText \"Tonight at 21.30: Music\", 3
<OK
```

CLEARPLAYLIST [<IMMEDIATELY>]

Clear the Ticker files playlist. If IMMEDIATELY is true the Ticker execution will be also interrupted.

Example:

```
>ClearPlaylist
<OK

>ClearPlaylist IMMEDIATELY=true
<OK
```

ADDPLAYLISTFILE <FILEPATH,MAXRECORDS_NUM,MAXRECORDS_FROMLAST>

Add the given file to the playlist at the last position. The file path may refer to either local or remote URL file. If MAXRECORDS_NUM is greater than 0, limit the number of records imported from the file. In this case MAXRECORDS_FROMLAST defines where the limit is applied: from the first or the last records.

Example:

```
>AddPlayListFile \"C://Datasource//tg.txt\"
<OK

>AddPlayListFile \"http://www.classx.it//prova.txt\"
<OK
```

ADDPLAYLISTFILES <FILEPATHS=<FILEPATH, MAXRECORDS_NUM, MAXRECORDS_FROMLAST>>

Add the given files to the playlist at the last position. The file path may refer to either local or remote URL file. If MAXRECORDS_NUM is greater than 0, limit the number of records imported from each file. In this case MAXRECORDS_FROMLAST defines where the limit is applied: from the first or the last records.

Example:

```
>AddPlaylistFiles
FILEPATHS=\"FILEPATH=\\\"C:\\\\\\\\\ClassX_Applications_v6\\\\\\\\shared\\\\\\\\datasource\\\\\\
\\class_cnbc\\\\\\\\tickers\\\\\\\\ita\\\\\\\\DAX30_INDEX\\\\\\\\2268719894846_Adidas
Ag.rss\\\"\\r\""
<RCVD: Ok
```

RESETPLAYLIST

Reset the playlist execution forcing the Ticker to restart by the first playlist file.

Example:

```
>ResetPlaylist
<OK
```

GETCURRENTTEXTDURATION

Return the scrolling time in milliseconds of the current text. The duration depends by the device resolution, the ticker width and the font size. Call this command while the text is scrolling.

Example:

```
>GetCurrentTextDuration
<21000
```

SETTICKERPROPERTY <SPEED> , <TEXTUPPERCASE>, <TEXTSEPARATOR>, <AUTOREMOVE>

Set the ticker properties. You can set one or more properties in the same command:

SPEED the ticker speed (px/s) [1, 30];

TEXTUPPERCASE true if convert to uppercase the ticker text;

TEXTSEPARATOR the ticker text separator;

AUTOREMOVE stop the ticker on finish the playlist execution;

Examples:

```
>SETSTYLEDTICKERPROPERTY "SPEED=8"
<OK

>SETSTYLEDTICKERPROPERTY "TEXTUPPERCASE=TRUE"
<OK

>SETSTYLEDTICKERPROPERTY "TEXTSEPARATOR=\" ~ \""
<OK

>SETSTYLEDTICKERPROPERTY "AUTOREMOVE=\"TRUE\""
<OK
```

8.5 Styled Ticker Horizontal

SETSTYLEDTICKERPROPERTY <SPEED> , <TEXTUPPERCASE>, <TEXTSEPARATOR>, <AUTOREMOVE>, <TEXTDIRECTIONMODE>

Set the ticker properties. You can set one or more properties in the same command:

SPEED the ticker speed (px/s) [1, 30];

TEXTUPPERCASE true if convert to uppercase the ticker text;

TEXTSEPARATOR the ticker text separator;

AUTOREMOVE stop the ticker on finish the playlist execution;

TEXTDIRECTIONMODE the ticker text direction [AUTOMATIC | LEFT_TO_RIGHT | RIGHT_TO_LEFT];

Note. AUTOMATIC: Automatically detect the text direction LTR or RTL

Examples:

```
>SETSTYLEDTICKERPROPERTY "TEXTDIRECTIONMODE=AUTOMATIC"
<OK

>SETSTYLEDTICKERPROPERTY "TEXTDIRECTIONMODE=LEFT_TO_RIGHT"
<OK

>SETSTYLEDTICKERPROPERTY "TEXTDIRECTIONMODE=RIGHT_TO_LEFT"
<OK
```

8.6 MediaText

SETTEXT <TEXT>

Set the text to display into the MediaText area.

Example:

```
>SetText \"Tonight 21.30: Blue Diamond\"
<OK
```

SETBACKGROUNDCOLOR <A>,<R>,<G>,<B>

Set the MediaText background color supporting also the alpha channel

Example:

```
>SetBackgroundColor A=255,R=255,G=0,B=0
<OK
```

SETFOREGROUNDCOLOR <A>,<R>,<G>,<B>

Set the MediaText foreground color supporting also the alpha channel.

Example:

```
>SetForegroundColor A=255,R=255,G=0,B=0
<OK
```

SETFOREGROUNDCOLOR <A>,<R>,<G>,<B>

Set the MediaText foreground color supporting also the alpha channel.

Example:

```
>SetForegroundColor A=255,R=255,G=0,B=0
<OK
```

8.7 TextSlideShow

SETFILENAME <FILE>

Set the TextSlideShow file to process and display.

Example:

```
>SetFileName \"C://Dataource/plugIn.csv\"
<OK
```

8.8 SimpleMedia

SETFILENAME <FILE>,<LOOPS>,<HQ>,<FILLALPHA>,<USEALPHA>

Set the SimpleMedia filename to play and enable loop and other execution properties. The loop parameter is not required and its default value is false. Use HQ to resize the media with best quality.

The default SimpleMedia screen location is (0,0) and its size is the same of the loaded file.

Example:

```
>SetFileName \"C:\\\\Video\\video_loop.avi\",LOOP,HQ
<OK

>SetFileName \"C:\\\\Video\\video_loop.avi\",HQ
<OK
```

SETMEDIAPROPERTY <LOOP>, <HQ>, <FILLALPHA>, <USEALPHA>, <SCALETOFIT>, <FIELDPOLARITY>, <SYNCHTOVIDEOREFRESH>, <AUDIOONLY>, <AUDIODELAY>, <VOLUME>, <PAUSEFIRST>, <PAUSELAST>, <MARKINMILLIS>, <MARKOUTMILLIS>, <MARKIN>, <MARKOUT>, <MEDIATIMEFRAME>, <MEDIATIMEMILLIS>, <PIXELASPECTRATIO>

Set the SimpleMedia properties. You can set one or more properties in the same command:

LOOP play the media in loop mode [TRUE | FALSE];

HQ use the best quality when scaling the media [TRUE | FALSE];

FILLALPHA fill the alpha channel of the media [TRUE | FALSE];

USEALPHA compose the alpha channel of the media with the background [TRUE | FALSE];

SCALETOFIT scale the media to fit the size assigned to the plugin [TRUE | FALSE];

FIELDPOLARITY **(DEPRECATED)** set the media video fields polarity [0 = EVEN, 1 = ODD, 2 = NONE];

FIELDFLAGS set the media video fieldflags mode [0 = NONE, 1 = DEINTERLACE, 2 = EVEN_DOMINANT, 3 = ODD_DOMINANT];

SYNCHTOVIDEOREFRESH synchronize the media with the output device video refresh freq [TRUE | FALSE];

AUDIOONLY use only the audio of the media (no video) [TRUE | FALSE];

AUDIODELAY set A/V delay;

VOLUME [0..255] set the volume of the audio.

PAUSEFIRST enable or disable the pause at first frame [TRUE | FALSE];

PAUSELAST enable or disable the pause at last frame [TRUE | FALSE];

MARKINMILLIS set the mark in [The media time in milliseconds];

MARKOUTMILLIS set the mark out [The media time in milliseconds];

MARKIN set the mark in [The media time in frames];

MARKOUT set the mark out [The media time in frames];

MEDIATIMEFRAME set the media time [The media time in frames];

MEDIATIMEMILLIS set the media time [The media time in milliseconds];

PIXELASPECTRATIO set the pixel aspect ratio from preset or ratio in floating value

[

FREE,

SQUARE,

PAL_NTSC_4_3,

PAL_NTSC_16_9,

PAL_4_3_DV,

PAL_16_9_DV,

NTSC_4_3_DV,

NTSC_16_9_DV

]

Example:

```
>SetMediaProperty LOOP
<OK
```

enable the play loop

```
>SetMediaProperty LOOP=FALSE,VOLUME=180
<OK
```

disable play loop and set clip volume to 180

```
>SetMediaProperty LOOP, HQ, FILLALPHA, USEALPHA, SCALETOFIT, FIELDPOLARITY=1,
SYNCHTOVIDEOREFRESH, AUDIOONLY, AUDIODELAY=600, VOLUME=180
<OK

>SetMediaProperty  MARKIN=120
<OK

>SetMediaProperty  MARKOUT=200
<OK

>SetMediaProperty  MARKINMILLIS=4200
<OK

>SetMediaProperty  MARKOUTMILLIS=6200
<OK

>SetMediaProperty  FIELDFLAGS=0
<OK
```

GETMEDIAPROPERTY <LOOP>, <HQ>, <FILLALPHA>, <USEALPHA>, <SCALETOFIT>, <FIELDPOLARITY>, <SYNCHTOVIDEOREFRESH>, <AUDIOONLY>, <AUDIODELAY>, <VOLUME>, <PAUSEFIRST>, <PAUSELAST>, <MARKINMILLIS>, <MARKOUTMILLIS>, <MARKIN>, <MARKOUT>, <MEDIATIMEFRAME>, <MEDIATIMEMILLIS>, <PIXELASPECTRATIO>, <DURATION>, <TOTFRAMES>, <ALL>

Get the SimpleMedia properties. You can get one or more properties in the same command:

LOOP get media loop mode [TRUE | FALSE];

HQ get media best quality flag [TRUE | FALSE];

FILLALPHA get the fill alha channel flag [TRUE | FALSE];

USEALPHA get the compose the alpha channel flag of the media [TRUE | FALSE];

SCALETOFIT get the scale to fit flag of the media size assigned to the plugin [TRUE | FALSE];

FIELDFLAGS get the media video fieldflags mode [0 = NONE, 1 = DEINTERLACE, 2 = EVEN_DOMINANT, 3 = ODD_DOMINANT];

SYNCHTOVIDEOREFRESH get the synchronize the media with the output device video refresh freq flag [TRUE | FALSE];

AUDIOONLY get the use only the audio of the media flag [TRUE | FALSE];

AUDIODELAY get A/V delay;

VOLUME [0..255] get the volume of the audio.

PAUSEFIRST get the the pause at first frame status [TRUE | FALSE];

PAUSELAST get the pause at last frame status [TRUE | FALSE];

MARKINMILLIS get the mark in [The media time in milliseconds];

MARKOUTMILLIS get the mark out [The media time in milliseconds];

MARKIN get the mark in [The media time in frames, -1 if no set];

MARKOUT get the mark out [The media time in frames, -1 if no set];

MEDIATIMEFRAME get the current media time [The media time in frames];

MEDIATIMEMILLIS get the current media time [The media time in milliseconds];

PIXELASPECTRATIO get the pixel aspect ratio floating value

DURATION get the media duration [The media duration in milliseconds];

TOTFRAMES get the media duration in frames [The media duration in frames];

ALL retrieve all properties;

Examples:

```
>GETMEDIAPROPERTY FIELDFLAGS
<FIELDFLAGS=0

>GETMEDIAPROPERTY FIELDFLAGS, LOOP, HQ, FILLALPHA, USEALPHA, SCALETOFIT, SYNCHTOVIDEOREFRESH,
AUDIOONLY, AUDIODELAY, VOLUME, MARKIN, MARKOUT, MARKINMILLIS, MARKOUTMILLIS, MEDIATIMEFRAME,
MEDIATIMEMILLIS, DURATION, TOTFRAMES
<FIELDFLAGS=0,LOOP="FALSE",HQ="TRUE",FILLALPHA="FALSE",USEALPHA="FALSE",SCALETOFIT="FALSE",SYNC
HTOVIDEOREFRESH="TRUE",AUDIOONLY="FALSE",AUDIODELAY=200,VOLUME=255,MARKIN=0,MARKOUT=0,MARKINMIL
LIS=-1,MARKOUTMILLIS=-1,              MEDIATIMEFRAME=0,MEDIATIMEMILLIS=0,PIXELASPECTRATIO=8.333,
DURATION=1000,TOTFRAMES=25

>GETMEDIAPROPERTY ALL
<FIELDFLAGS=0,LOOP="FALSE",HQ="TRUE",FILLALPHA="FALSE",USEALPHA="FALSE",SCALETOFIT="FALSE",SYNC
HTOVIDEOREFRESH="TRUE",AUDIOONLY="FALSE",AUDIODELAY=200,VOLUME=255,MARKIN=0,MARKOUT=0,MARKINMIL
LIS=-1,MARKOUTMILLIS=-
1,MEDIATIMEFRAME=0,MEDIATIMEMILLIS=0,PIXELASPECTRATIO=8.333,DURATION=1000,TOTFRAMES=25
```

ADDFRAMEMARKER <FRAME>, <TYPE>, <LOOPS>

Add a FrameMarker at a specific frame.

FRAME frame number for adding the FrameMarker

TYPE FrameMarker type, it can assume one of the following values: TRIGGER_AND_PAUSE, TRIGGER_AND_CONTINUE, TRIGGER_AND_LOOP_START, TRIGGER_AND_LOOP_END, TRIGGER_AND_MARKIN, TRIGGER_AND_MARKOUT

LOOPS number of loops. It is valid only if the FrameMarker type is: TRIGGER_AND_LOOP_END

Examples:

```
>LBC_SENDCONTENTCOMMAND "SIGLA", "ADDFRAMEMARKER 185, TRIGGER_AND_CONTINUE"
<Ok

>LBC_SENDCONTENTCOMMAND "SIGLA", "ADDFRAMEMARKER 268, TRIGGER_AND_LOOP_END, 3"
<Ok
```

REMOVEFRAMEMARKER <FRAME>

Remove the FrameMarker at a specific frame.

FRAME frame number for deleting the FrameMarker

Examples:

```
>LBC_SENDCONTENTCOMMAND "SIGLA", "REMOVEFRAMEMARKER 185"
<Ok
```

GETFRAMEMARKERS

Return the list of the FrameMarker .

Examples:

```
>LBC_SENDCONTENTCOMMAND "SIGLA", "GETFRAMEMARKERS"
<FRAME=38,TYPE="TRIGGER_AND_MARKIN"
 FRAME=130,TYPE="TRIGGER_AND_LOOP_START"
 FRAME=240,TYPE="TRIGGER_AND_CONTINUE"
 FRAME=417,TYPE="TRIGGER_AND_PAUSE"
 FRAME=1583,TYPE="TRIGGER_AND_LOOP_END", LOOPS=3
```

```
FRAME=1696,TYPE="TRIGGER_AND_MARKOUT"
```

SETCROPINFO <ENABLE>, <PARAM0>, <PARAM1>, <PARAM2>, <PARAM3>

Set the crop info. You can set one or more properties in the same command:

ENABLE set the crop enable [TRUE | FALSE];

PARAM0
PARAM1
PARAM2
PARAM3 are the crop integer values in pixel;

This command respects the HTML#margins standard.

If you set 4 values are applied with order top/right/bottom/left
If you set 3 values are applied with order top/left/bottom
If you set 2 values the first value is applied to top/bottom and the second is applied to left/right
If you set 1 values are applied in all sides

Examples:

```
// ENABLE="TRUE",TOP=10,RIGHT=11,BOTTOM=12,LEFT=13
>SETCROPINFO ENABLE 10 11 12 13
<OK

// ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=13
>SETCROPINFO ENABLE 10 11 12
<OK

// ENABLE="TRUE",TOP=15,RIGHT=20,BOTTOM=15,LEFT=20
>SETCROPINFO ENABLE 15 20
<OK

// ENABLE="TRUE",TOP=10,RIGHT=10,BOTTOM=10,LEFT=10
>SETCROPINFO ENABLE 10
<OK
```

GETCROPINFO <ENABLE>, <TOP>, <RIGHT>, <BOTTOM>, <LEFT>, <ALL>

Set the crop info. You can set one or more properties in the same command:

ENABLE set the crop enable [TRUE | FALSE];

TOP get the crop top integer value in pixel;
RIGHT get the crop right integer value in pixel;
BOTTOM get the crop bottom integer value in pixel;
LEFT get the crop left integer value in pixel;

ALL retrieve all properties;

Examples:

```
>LBC_SENDCONTENTCOMMAND "SIGLA", "GETCROPINFO ENABLE TOP RIGHT BOTTOM LEFT"
<ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=11

>LBC_SENDCONTENTCOMMAND "SIGLA", "GETCROPINFO ALL"
<ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=11
```

8.9 Image

SETFILENAME <FILE>,<USEALPHA>,<HQ>

Set the Background filename to display. The USEALPHA parameter is not required and its default value is false. Use HQ to resize the media with best quality.

The default Background screen location is (0,0) and its size is the same of the loaded file.

Example:

```
>SetFileName \"C:\\\\Video\\video_loop.avi",ALPHA,HQ
<OK

>SetFileName \"C://Video/video_loop.avi",HQ
<OK
```

8.10 MediaPlayer

CLEARPLAYLIST

Clear the MediaPlayer files playlist.

Example:

```
>ClearPlaylist
<OK
```

ADDPLAYLISTFILE <FILEPATH>

Add the given file to the playlist at the last position. The file path may refer to either local or remote URL file.

Example:

```
>AddPlayListFile \"C://Datasource//tg.txt\"
<OK

>AddPlayListFile \"http://www.classx.it//prova.txt\"
<OK
```

RESETPLAYLIST

Reset the playlist execution forcing the MediaPlayer to restart by the first playlist file.

Example:

```
>ResetPlaylist
<OK
```

SETMEDIAPROPERTY <LOOP>, <HQ>, <FILLALPHA>, <USEALPHA>, <SCALETOFIT>, <FIELDPOLARITY>, <SYNCHTOVIDEOREFRESH>, <AUDIODELAY>, <VOLUME>

Set the SimpleMedia properties. You can set one or more properties in the same command:

LOOP play the media in loop mode;

HQ use the best quality when scaling the media;

FILLALPHA fill the alpha channel of the media;

USEALPHA compose the alpha channel of the media with the background;

SCALETOFIT scale the media to fit the size assigned to the plugin;

FIELDPOLARITY **(DEPRECATED)** set the media video fields polarity [0 = EVEN, 1 = ODD, 2 = NONE];

FIELDFLAGS set the media video fieldflags mode [0 = NONE, 1 = DEINTERLACE, 2 =

EVEN_DOMINANT, 3 = ODD_DOMINANT];

SYNCHTOVIDEOREFRESH synchronize the media with the output device video refresh freq;

AUDIODELAY set A/V delay;

VOLUME [0..255] set the volume of the audio.

Example:

```
>SetMediaProperty LOOP
<OK
```

enable the play loop

```
>SetMediaProperty LOOP=FALSE,VOLUME=180
<OK
```

disable play loop and set clip volume to 180

```
>SetMediaProperty LOOP, HQ, FILLALPHA, USEALPHA, SCALETOFIT, FIELDPOLARITY=1,
SYNCHTOVIDEOREFRESH, AUDIODELAY=600, VOLUME=180
<OK
```

SETCROPINFO <ENABLE>, <PARAM0>, <PARAM1>, <PARAM2>, <PARAM3>

Set the crop info. You can set one or more properties in the same command:

ENABLE set the crop enable [TRUE | FALSE];

PARAM0
PARAM1
PARAM2
PARAM3 are the crop integer values in pixel;

This command respects the HTML#margins standard.

If you set 4 values are applied with order top/right/bottom/left
If you set 3 values are applied with order top/left/bottom
If you set 2 values the first value is applied to top/bottom and the second is applied to left/right
If you set 1 values are applied in all sides

Examples:

```
// ENABLE="TRUE",TOP=10,RIGHT=11,BOTTOM=12,LEFT=13
>SETCROPINFO ENABLE 10 11 12 13
<OK

// ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=13
>SETCROPINFO ENABLE 10 11 12
<OK

// ENABLE="TRUE",TOP=15,RIGHT=20,BOTTOM=15,LEFT=20
>SETCROPINFO ENABLE 15 20
<OK

// ENABLE="TRUE",TOP=10,RIGHT=10,BOTTOM=10,LEFT=10
>SETCROPINFO ENABLE 10
<OK
```

GETCROPINFO <ENABLE>, <TOP>, <RIGHT>, <BOTTOM>, <LEFT>, <ALL>

Set the crop info. You can set one or more properties in the same command:

ENABLE set the crop enable [TRUE | FALSE];

TOP get the crop top integer value in pixel;
RIGHT get the crop right integer value in pixel;
BOTTOM get the crop bottom integer value in pixel;
LEFT get the crop left integer value in pixel;

ALL retrieve all properties;

Examples:

```
>LBC_SENDCONTENTCOMMAND "SIGLA", "GETCROPINFO ENABLE TOP RIGHT BOTTOM LEFT"
<ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=11

>LBC_SENDCONTENTCOMMAND "SIGLA", "GETCROPINFO ALL"
<ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=11
```

8.11 SlideShow

CLEARPLAYLIST

Clear the Slideshow files playlist.

Example:

```
>ClearPlaylist
<OK
```

ADDPLAYLISTFILE <FILEPATH>

Add the given file to the playlist at the last position. The file path may refer to either local or remote URL file.

Example:

```
>AddPlayListFile \"C://Datasource//tg.txt\"
<OK

>AddPlayListFile \"http://www.classx.it//prova.txt\"
<OK
```

RESETPLAYLIST

Reset the playlist execution forcing the Slideshow to restart by the first playlist file.

Example:

```
>ResetPlaylist
<OK
```

8.12 NDI Source

SETCROPINFO <ENABLE>, <PARAM0>, <PARAM1>, <PARAM2>, <PARAM3>

Set the crop info. You can set one or more properties in the same command:

ENABLE set the crop enable [TRUE | FALSE];

PARAM0
PARAM1
PARAM2
PARAM3 are the crop integer values in pixel;

This command respects the HTML#margins standard.

If you set 4 values are applied with order top/right/bottom/left
If you set 3 values are applied with order top/left/bottom
If you set 2 values the first value is applied to top/bottom and the second is applied to left/right
If you set 1 values are applied in all sides

Examples:

```
// ENABLE="TRUE",TOP=10,RIGHT=11,BOTTOM=12,LEFT=13
>SETCROPINFO ENABLE 10 11 12 13
<OK
```

```
// ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=13
>SETCROPINFO ENABLE 10 11 12
<OK

// ENABLE="TRUE",TOP=15,RIGHT=20,BOTTOM=15,LEFT=20
>SETCROPINFO ENABLE 15 20
<OK

// ENABLE="TRUE",TOP=10,RIGHT=10,BOTTOM=10,LEFT=10
>SETCROPINFO ENABLE 10
<OK
```

GETCROPINFO <ENABLE>, <TOP>, <RIGHT>, <BOTTOM>, <LEFT>, <ALL>

Set the crop info. You can set one or more properties in the same command:

ENABLE set the crop enable [TRUE | FALSE];

TOP get the crop top integer value in pixel;
RIGHT get the crop right integer value in pixel;
BOTTOM get the crop bottom integer value in pixel;
LEFT get the crop left integer value in pixel;

ALL retrieve all properties;

Examples:

```
>LBC_SENDCONTENTCOMMAND "SIGLA", "GETCROPINFO ENABLE TOP RIGHT BOTTOM LEFT"
<ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=11

>LBC_SENDCONTENTCOMMAND "SIGLA", "GETCROPINFO ALL"
<ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=11
```

8.13 SharedFrame

SETCROPINFO <ENABLE>, <PARAM0>, <PARAM1>, <PARAM2>, <PARAM3>

Set the crop info. You can set one or more properties in the same command:

ENABLE set the crop enable [TRUE | FALSE];

PARAM0
PARAM1
PARAM2
PARAM3 are the crop integer values in pixel;

This command respects the HTML#margins standard.

If you set 4 values are applied with order top/right/bottom/left
If you set 3 values are applied with order top/left/bottom
If you set 2 values the first value is applied to top/bottom and the second is applied to left/right
If you set 1 values are applied in all sides

Examples:

```
// ENABLE="TRUE",TOP=10,RIGHT=11,BOTTOM=12,LEFT=13
>SETCROPINFO ENABLE 10 11 12 13
<OK
```

```
// ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=13
>SETCROPINFO ENABLE 10 11 12
<OK

// ENABLE="TRUE",TOP=15,RIGHT=20,BOTTOM=15,LEFT=20
>SETCROPINFO ENABLE 15 20
<OK

// ENABLE="TRUE",TOP=10,RIGHT=10,BOTTOM=10,LEFT=10
>SETCROPINFO ENABLE 10
<OK
```

GETCROPINFO <ENABLE>, <TOP>, <RIGHT>, <BOTTOM>, <LEFT>, <ALL>

Set the crop info. You can set one or more properties in the same command:

ENABLE set the crop enable [TRUE | FALSE];

TOP get the crop top integer value in pixel;
RIGHT get the crop right integer value in pixel;
BOTTOM get the crop bottom integer value in pixel;
LEFT get the crop left integer value in pixel;

ALL retrieve all properties;

Examples:

```
>LBC_SENDCONTENTCOMMAND "SIGLA", "GETCROPINFO ENABLE TOP RIGHT BOTTOM LEFT"
<ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=11

>LBC_SENDCONTENTCOMMAND "SIGLA", "GETCROPINFO ALL"
<ENABLE="TRUE",TOP=10,RIGHT=0,BOTTOM=12,LEFT=11
```

8.14 Video Input

SETSOURCEID <SOURCEID>

It changes the ID number of the source on the Video Input

SOURCEID ID number

Example:

```
>LBC_SENDCONTENTCOMMAND "contentName", "SetSourceID 2"
<OK
```

SETVOLUME <SETVOLUME>

Sets the volume to the specified video input source.

SETVOLUME percentage of volume

Example:

```
>LBC_SENDCONTENTCOMMAND "contentName", "SetVolume 10"
```

```
<OK
```

NB: this method does not modify in any way the master volume of the source. As soon as the master volume is reset, this volume is reset. (the values of the volume can be viewed on the tooltips of the slider of the VideoInput preview panel). This volume is relatively calculated compared to the master volume.

9 LiveBoardSCG extension

ClassX OEM products suite contains the special application LiveBoardSCG (LiveBoard for StealthCG) that integrates the flexibility of LiveBoard to manage live events with the graphics produced by a video system automation through StealthCG. For instance while StealthCG is playing a ticker and a logo under control of the automation, an user can create and play a graphic content at the same time, using the same output device.

The existing contents of LiveBoard may also be controlled using specific TCP/IP commands, that allows to play, stop and edit each content. It is not possible to create and destroy a content through remote commands. This means that only already existing contents may be controlled. The user for instance, can create contents manually or loading a LiveBoard project.

The interface that allows to control the contents of LiveBoard remotely is named LBControl. LBControl exposes both an API for scripting purposes through the built-in scripting engine, and TCP/IP protocol that emulates the API functionality. The following section contains a list of all supported commands: a content is identified uniquely by its name. If there are more than one contents using the same name, it not possible to know which content will be controlled.

LBC_GETPROTOCOLVERSION

Return the protocol version

LBC_PLAYCONTENT <CONTENTNAME>

Play the content with the given CONTENTNAME.

LBC_PLAYCONTENT <CONTENTNAMELIST>

Play the given contents list synchronously. The list must follow the syntax: CONTENTNAMELIST="NAME1,NAME2,NAME3".
If one or more contents name in the list is not valid, the command will not be executed.

Example: play the content news,digital clock,ticker at the same time.

```
>LBC_PLAYCONTENT "news,digital clock,ticker"

or

>LBC_PLAYCONTENT CONTENTNAMELIST="news,digital clock,ticker"
```

LBC_STOPCONTENT <CONTENTNAME>

Stop the content with the given CONTENTNAME.

LBC_STOPALL

Stop all playing contents.

LBC_STOPCONTENTS <CONTENTNAMELIST>

Stop the given contents list synchronously. The list must follow the syntax: CONTENTNAMELIST="NAME1,NAME2,NAME3".
If one or more content name in the list is not valid, the command will not be executed.

Example: stop the content news,digital clock,ticker at the same time.

```
>LBC_STOPCONTENTS "news,digital clock,ticker"

or

>LBC_STOPCONTENTS CONTENTNAMELIST="news,digital clock,ticker"
```

LBC_SENDCONTENTCOMMAND <CONTENTNAME>,<COMMAND>

Send a command to the content with the given CONTENTNAME.
Each content can have a set of custom commands to control many internal aspects, behaviours and settings. Normally only the MediaPlugs expose this feature. For more information please consult the **MediaPlugs Commands** section.

To obtain a correct interpretation of the command parameter, it should be written by replacing special characters with the equivalent ANSI-C escape sequence. See the **Special Characters** table.

Example: change the image file of a Background plugin.

```
>LBC_SENDCONTENTCOMMAND CONTENTNAME="BGND",COMMAND="SetFilename \"C:\\\\Desktop\\\\image.jpg\""
```

LBC_PAUSECONTENT <CONTENTNAME>

Pause the content with the given CONTENTNAME.
NOTE: Not all contents may be paused.

LBC_RESUMECONTENT <CONTENTNAMELIST>

Resume the contents in list CONTENTNAMELIST="NAME1,NAME2,NAME3".
If one or more content name in the list is not valid, the command will not be executed.

NOTE: Only paused content may be resumed.

LBC_RESUMECONTENTS <CONTENTNAME>

Resume the content with the given CONTENTNAME.
NOTE: Only paused content may be resumed.

LBC_GETCONTENTKIND<CONTENTNAME>

Return the kind of the Conent with the given name. It may be IMAGE | GROUP | MEDIA | TEMPLATE | GROUP | SCRIPT | PLUGIN.

LBC_SETCONTENTBOUNDS <CONTENTNAME>,<X>,<Y>,<WIDTH>,<HEIGHT>,<UPDATEGUI>

Set the CONTENTNAME content bounds. All parameters < 0 will be ignored.
UPDATEGUI true forces the repaint of LiveBoard interface. Be careful to use this feature because it may interfere with correct execution of LiveBoard.

NOTE: Not all contents support this command.

LBC_SETCONTENTDESCRIPTION <CONTENTNAME>,<DESCRIPTION>

Set the description of the given CONTENTNAME.

LBC_SETTEMPLATEOBJECTVALUE <CONTENTNAME>,<OBJECTNAME>,<VALUE>

Allows to modify the value of the given editable object name contained into the graphic template at the given CONTENTNAME. This command can be applied to all objects kind and the result depends on their implementation. For example this command changes the text value for a TEXT object and also set the file path for an IMAGE or VIDEO object.

NOTE: LBControl does not check the correctness of the value. It means that if the file path of an image is wrong LBControl does not return any error, but the IMAGE object will display an empty image.

Example

```
>LBC_SETTEMPLATEOBJECTVALUE "graphic_template","news","name","Adam"
<OK

> LBC_SETTEMPLATEOBJECTVALUE "graphic_template","soccer","team","gfx/milan.tga"
<OK

> LBC_SETTEMPLATEOBJECTVALUE "graphic_template","soccer","logo","gfx/milan_logo.avi"
<OK
```

LBC_SETCONTENTVISUALPRIORITY <CONTENTNAME>,<VALUE>

This command allows to change the visibility order of the content: 0 is the lower layer and 9999 is the higher layer.

LBC_TEMPLATENEXTRECORD <CONTENTNAME>,<DATASOURCEID>,<STEPS>

This command allows to change the RecordID of some DataSources assigned to the CONTENTNAME. In particularly this moves the RecordID to the next record about STEPS. If steps is not specified it is considered equal to 1. When the next record reaches the last, it will restart from the beginning automatically. The DataSourceID identifies one or more DataSources

LBC_TEMPLATEPREVIOUSRECORD <CONTENTNAME>,<DATASOURCEID>,<STEPS>

This command allows to change the RecordID of some DataSources assigned to the CONTENTNAME. In particularly this moves the RecordID to the previous record about STEPS. If steps is not specified it is considered equal to 1. When the next record reaches the first, it will restart from the beginning automatically. The DataSourceID identifies one or more DataSources

LBC_TEMPLATESETRECORD <CONTENTNAME>,<DATASOURCEID>,<RECORD>

This command allows to change the RecordID of some DataSources assigned to the CONTENTNAME. In particularly this moves the RecordID to the given record position. The record is clamped to the DataSource records number. The DataSourceID identifies one or more DataSources.

LBC_TEMPLATEDATASOURCERELOAD<CONTENTNAME>,<DATASOURCEID>

This command reload all DataSources to the CONTENTNAME. In particularly the DataSourceID identifies one or more DataSources.

LBC_TEMPLATEEXITFROMLOOP<CONTENTNAME>

An LBEngineContent may loop in a FrameMarker range. The loops number may be fixed or infinite. In both cases this command force the LBEngineContent to exit from the loop.

10 OutputDevice Remote Protocol

The Output device protocol collects a set of remote commands which allow to change some settings of the Output devices used by ClassX Applications. This document does not contain all remote protocol specifications and how to, but must be considered as an **add-on** to the main document.

This protocol changes the device settings whose names are collected into an enumeration named PARAMS. Any PARAM changes will have effect when the UPDATEFLDPARAMS command is called. Each parameter is described by a name that is unique.

The PARAMS supported by the protocol

```
FIELDMODE_PSF          // the psf video format

// PIP1 creation/update params
PIP1_VISIBLE           // 0 is invisible
PIP1_X                 // x coord
PIP1_Y                 // y coord
PIP1_WIDTH             // width
PIP1_HEIGHT            // height
PIP1_HIGHQUALITY       // 0 is low quality
PIP1_VOLUME            // audio volume [0 to 100]
PIP1_ALPHA             // alpha channel transparence [0 to 100]

PIP1_CROPLEFT          // left crop pixels
PIP1_CROPRIGHT         // right crop pixels
PIP1_CROPTOP           // top crop pixels
PIP1_CROPBOTTOM        // bottom crop pixels
PIP1_CROPENABLE        // crop enable (0 is disable)

// global pip params
PIP_OVERGRAPHICS       // 0 is under graphics (compositing), else is over graphics layer
PIP_ANIMSPEED          // 0 is immediate, else is the fps of the interpolation animation between
different PIP positions
PIP_ANIMRAMP           // 0 is motion linear, else is accelerated
```

FLD_GETPARAMS

Return a list of the device PARAMS. Each line of the list contains the parameter name and its current value in the format:

PARAM=<param>, VALUE="<value>".

Example:

```
-> FLD_GETPARAMS
<- PARAM=PIP_VISIBLE, VALUE="1"
PARAM=PIP0_X, VALUE="25"
PARAM=PIP0_Y, VALUE="65"
PARAM=PIP0_WIDTH, VALUE="39"
PARAM=PIP0_HEIGHT, VALUE="29"
PARAM=PIP0_HIGHQUALITY, VALUE="0"
```

FLD_GETPARAM[PARAM]

Return the current value of the given parameter.

Example:

```
-> FLD_GETPARAM PARAM=PIP_X
<- PARAM=PIP_X, VALUE="25"
```

FLD_SETPARAM[PARAM,VALUE]

Set the value into the parameter. Please remember that changes will be applied to the driver when the

UPDATEFLDPARAMS is called.

Example:

```
-> FLD_SETPARAM PARAM=PIP_Y VALUE="55"
<- OK
```

FLD_UPDATEPARAMS

Apply the current params values to the Output device. This call may require time when the device has to be reinitialized. Not at all params require the device reinitialization.

11 Error management

GETLASTERRORMESSAGE

Return the description of the last found error.

```
>GETLASTERRORMESSAGE
<UNKNOWN COMMAND
```

GETERRORMESSAGE <ERRORCODE>

Return the error message description according with the given ERRORCODE.

```
>ERRORMESSAGE 4
<INVALID FILE NAME
```

GETLASTERROR

Return the last found error.

```
>GETLASTERROR
<ERROR 2
```

SHOWDIALOGMESSAGES <ENABLE>

Disable warning or error messages displaying

```
>SHOWDIALOGMESSAGES ENABLE=FALSE
<OK
```

11.1 Error table

```
// Common errors
ERR_NONE                        =0,"No errors"
ERR_UNKNOWN                     =1,"Unknown error"
ERR_UNKNOWNCOMMAND              =2,"Unknown command"
ERR_INVALIDFILENAME             =3,"Invalid file name"
ERR_BADPARAMETERS               =4,"Bad parameters"
ERR_PLUGINCOMMANDFAILED         =5,"Plugin command failed"
ERR_NOTIMPLEMENTED              =6,"Command not implemented"
ERR_SHOULDQUIT                  =7,"Quit"
ERR_PARSE                       =8,"Command parsing error"
ERR_FILEDOESNTEXIST             =9,"The file does not exist"
ERR_FILEALREADYEXIST            =10,"The file already exist"

// CGTemplateEditor/CGPlayout Errors
CGPE_ERR_BASE                   =1000,"Error base for CGPlayout and CGTemplateEditor"
CGPE_ERR_LOADPROJECT            =1001,"Project load error"
CGPE_ERR_INVALIDENGINE          =1002,"Invalid engine"
CGPE_ERR_INVALIDINDEX           =1003,"Invalid project index"
CGPE_ERR_PROJECTNOTFOUND        =1004,"Project not found"
CGPE_ERR_COMMANDFORBIDDEN       =1005,"Command forbidden while playing"
CGPE_ERR_INVALIDFRAME           =1006,"Frame not valid"
CGPE_ERR_OBJECTNOTFOUND         =1007,"Object not found"
```

```
CGPE_ERR_NODATASOURCES              =1008,"No data sources available"
CGPE_ERR_NOLOADEDPROJECTS           =1009,"No loaded projects"
CGPE_ERR_MEDIAFILENOTFOUND          =1010,"Video or audio file not found"
CGPE_ERR_NOLOADEDPLUGINS            =1011,"No plugins loaded"
CGPE_ERR_NOEDITABLEOBJECTS          =1012,"No editable objects"
CGPE_ERR_INVALIDENGINEID            =1013,"Invalid engine ID, it is null or contains invalid chars or
                                           is already assigned"
CGPE_ERR_TIMEOUT                    =1014,"Timeout error waiting the plugin status"
CGPE_ERR_DEVICENOTINITIALIZED       =1015,"The device is not initialized"

// StealthCG errors
SCG_ERR_BASE                        =2000,"Error base for StealthCG"
SCG_ERR_INVALIDCONTENTID            =2001,"Invalid content ID, it is null or contains invalid chars"
SCG_ERR_CONTENTNOTFOUND             =2002,"Content not found"
SCG_ERR_PLUGINTYPE                  =2003,"Unknown plugIn type"
SCG_ERR_INVALIDCONTENTOPERATION     =2004,"The operation is not supported by the content"
SCG_ERR_INVALIDTEMPLATEOBJECT       =2005,"Cannot find the template object to edit"
SCG_ERR_NOCONTENTCOMMANDS           =2006,"No commands for the content"
SCG_ERR_CONTENTALREADYEXISTS        =2007,"A content with the same ID already exists"
SCG_ERR_INVALIDTEMPLATE             =2008,"The template contains invalid data"
SCG_ERR_SAVEPROJECTFAILED           =2009,"Save project failed"
SCG_ERR_DEVICENOTINITIALIZED        =2010,"The device is not initialized"

// LBControl erros
LBC_ERR_BASE                        =3000,"Error base for LBControl"
LBC_ERR_CONTENTNOTFOUND             =3001,"Content not found"
LBC_ERR_INVALIDCONTENT              =3002,"Invalid content for the operation"
LBC_ERR_INVALIDTEMPLATEOBJECT       =3003,"Cannot find the template object to edit"
LBC_ERR_NOCONTENTCOMMANDS           =3004,"No commands for the content"
LBC_ERR_TEMPLATEOBJECTNOTFOUND      =3005,"Template object not found"

// Social Server errors
SCS_ERR_NOMESSAGES                  =4000,"No messages"
```

12 Change log

Version 1.3.1

- [new] Support OutputDeviceRemote protocol to control and setup the application output device

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